# Drift or Return? Fuel-Optimal Recovery of Perturbed Along-Track Satellite Formations via Sequential Convex Programming


Rahul Ayanampudi
AeroAstro Department
Stanford University
rayanam@stanford.edu

Zeb Barry
AeroAstro Department
Stanford University
zebbarry@stanford.edu

Julian Brown
AeroAstro Department
Stanford University
julianb@stanford.edu

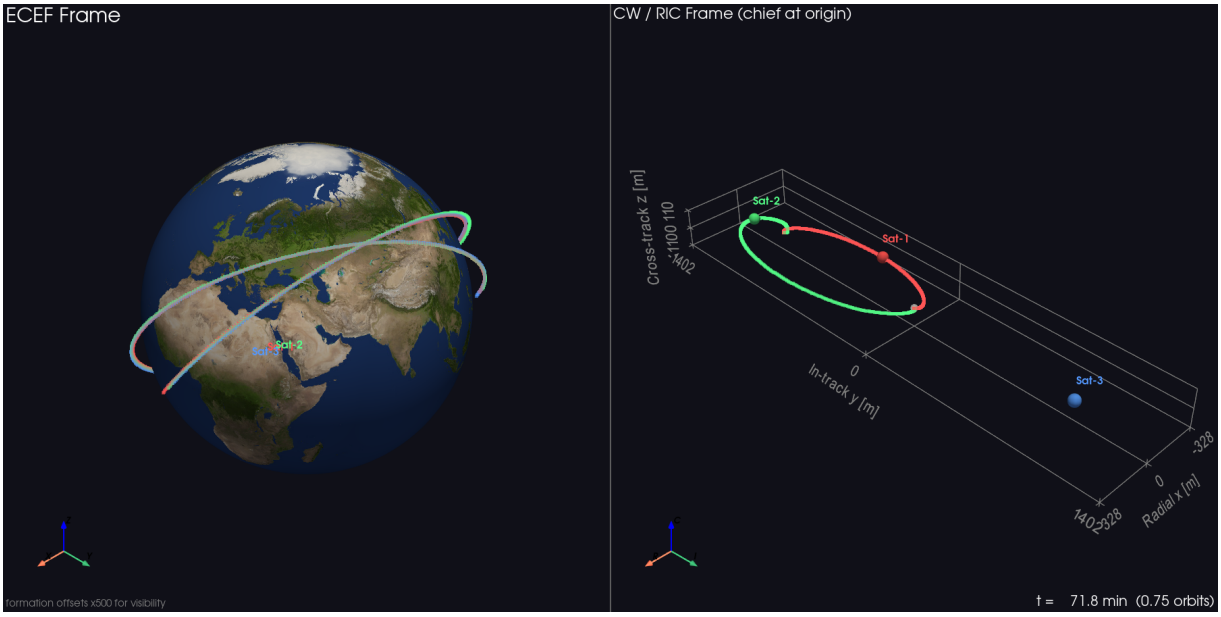




Figure 1: Dual-frame visualization of an SCP formation recovery. Left: Earth-Centered Earth-Fixed (ECEF) view of the 550 km, 51.6° chief reference orbit (a regressing spiral in the rotating frame) with the three labeled satellites; formation offsets are exaggerated 500× so the satellites are individually visible at planetary scale. Right: Clohessy–Wiltshire (CW) / RIC view centered on the virtual chief: after an unexpected $\delta\mathbf{v}_0$ on Sat-2 (green), the SCP planner elects to swap Sat-2 and Sat-1 (red) rather than fight the natural along-track drift, restoring the original 1 km spacing for 0.30 m/s of total $\Delta v$ versus 2.27 m/s for the return-to-slot baseline, while Sat-3 (blue) holds station. The simulator, solvers, and all figures are produced by our open Python package cwsim.


## Abstract

Close-proximity satellite formations in Low Earth Orbit (LEO) operate with narrow safety margins: a single unexpected velocity perturbation—a collision avoidance burn or thruster misfire—induces secular along-track drift that carries the perturbed satellite across the trajectories of its neighbors. We ask whether the optimal response is to return the perturbed satellite to its original slot, or to exploit the natural drift and reorder the formation. We model a three-satellite along-track formation with the full three-degree-of-freedom Clohessy–Wiltshire (CW) dynamics, propagated exactly through the closed-form state transition matrix (STM), and compare two recovery architectures: (a) a closed-form two-impulse CW solver that restores the perturbed satellite to its original slot, and (b) a Sequential Convex Programming (SCP) planner that co-

ordinates impulsive burns across all satellites, permits any slot ordering that reproduces the original spacing, and enforces pairwise keep-out spheres, per-burn thrust limits, and per-satellite fuel budgets. Because the CW dynamics are exactly linear, the keep-out constraint is the only non-convexity; we linearize it as the supporting halfspace of the keep-out sphere, a conservative inner approximation that renders every SCP iterate collision-free at every imposed sample—and because the inter-node coasting state is affine in the decision variables, the constraint is imposed densely in time along the whole trajectory, not only at burn nodes. Across a sweep of perturbation directions, magnitudes, and reaction delays, SCP reduces total constellation $\Delta v$ by up to 89% (median ≈56%) relative to the baseline; for along-track perturbations the optimizer always chooses to re-slot, confirming that leveraging the drift is cheaper than reversing it. We further show that the baseline, which carries no collision logic, can violate a 200 m keep-out that the SCP solution honors, and that its cost diverges when the transfer horizon approaches half-period multiples.

## Abbreviations

CW: Clohessy–Wiltshire; ECEF: Earth-Centered Earth-Fixed; ECI: Earth-Centered Inertial; LEO: Low Earth Orbit; RIC: Radial–In-track–Cross-track; RK4: fourth-order Runge–Kutta; SCP: Sequential Convex Programming; SOCP: Second-Order Cone Program; STM: State Transition Matrix.

## 1. Introduction

Close-proximity satellite formations enable distributed sensing, interferometry, and on-orbit servicing in LEO, exemplified by the NASA Starling swarm demonstration [6], but they operate with narrow safety margins. A single spacecraft can experience an unexpected orbital perturbation—a collision avoidance burn, a thruster misfire—which introduces secular along-track drift and Coriolis coupling that rapidly drive the perturbed satellite out of formation and across the trajectories of its neighbors [3]. The operational question we study is whether it is best to return the perturbed satellite to its original slot using the closed-form CW solution [1], or whether the natural drift can be leveraged: a coordinated re-ordering of the formation that reduces total constellation fuel consumption and thereby extends mission life.

Input and output: The input to our planner is the post-perturbation state of the constellation—each satellite's RIC relative state $\mathbf{x}^{(i)} = [x, y, z, \dot{x}, \dot{y}, \dot{z}]^\top \in \mathbb{R}^6$—together with a reaction delay $t_d$ (no burns permitted before $t_d$) and a recovery time $t_f$. The output is a complete impulsive burn schedule $\{(t_k, \Delta\mathbf{v}_k^{(i)})\}$ for every satellite and a slot assignment $\sigma$ such that at $t_f$ the constellation occupies the original equally spaced along-track geometry in some permutation, subject to pairwise keep-out spheres, per-burn thrust limits, and per-satellite fuel budgets.

The key design choices are (i) to let the solver plan burns for all satellites, not only the perturbed one, so that a cheap cooperative swap can replace an expensive unilateral return; (ii) to treat the slot assignment as a discrete outer decision ranked by a closed-form two-impulse cost heuristic; and (iii) to exploit the exact linearity of the CW dynamics so that the pairwise collision constraint is the only non-convexity handled by successive linearization.

Our contribution is fourfold: (i) cwsim, a modular Python simulator built on the exact closed-form 3-DOF CW STM, with impulsive maneuvers, analytic ECI chief propagation, and an interactive dual-frame (ECEF + RIC) 3D visualization; (ii) a closed-form two-impulse CW baseline controller; (iii) an SCP recovery planner with slot re-assignment, keep-out spheres, thrust and fuel limits, fuel balancing across the constellation, and an outer time-of-flight trade; and (iv) a trade study over perturbation direction, magnitude, and reaction delay that quantifies when re-slotting beats return-to-slot.

## 2. Related Work

Linearized relative orbital motion dates to Hill's lunar-theory equations [2]; Clohessy and Wiltshire [1] specialized them to satellite terminal rendezvous about a circular reference orbit, yielding the closed-form solution and two-impulse targeting law we use as our baseline. Standard treatments of the dynamics, the STM, and the half-period targeting singularity appear in Vallado [3] and Schaub and Junkins [4]; Alfriend et al. [5] survey formation flying dynamics and control, including impulsive formation reconfiguration and the along-track equilibrium family our slots are drawn from.

Convex optimization has become a workhorse for spacecraft trajectory planning because interior-point methods deliver certifiable optima in polynomial time [11]. Açıkmeşe and Ploen [7] convexified powered-descent guidance; Lu and Liu [8] posed rendezvous and proximity operations as a sequence of SOCPs. Closest to our setting, Morgan et al. [9] planned fuel-optimal swarm reconfiguration with SCP, linearizing the nonconvex inter-agent collision constraint as a supporting halfspace—the same conservative inner approximation we adopt—and Mao et al. [10] analyze convergence of

successive convexification. Our problem differs in emphasis: the dynamics are exactly linear, so no trust region is needed for dynamic feasibility, and we couple the continuous burn optimization with a discrete slot-assignment search, benchmarked directly against the closed-form two-impulse law. The implementation uses CVXPY [12] with the Clarabel interior-point solver [13], and PyVista [14] for 3D visualization.

## 3. Methods

### 3.1. Relative dynamics and the 3-DOF STM

We use the chief-centered radial–in-track–cross-track (RIC, or Hill) frame with $x$ radial (zenith), $y$ in-track (along the chief velocity), and $z$ cross-track (orbit normal). Some references call this frame LVLH loosely; the strict LVLH convention places $z$ toward nadir and $x$ along velocity, and differs from RIC by a proper rotation. We take a circular chief orbit of radius $a_0 = R_\oplus + 550\,\mathrm{km}$ with mean motion $n = \sqrt{\mu/a_0^3}$. The CW equations are

$$\ddot{x} = 3n^2 x + 2n\dot{y}, \qquad \ddot{y} = -2n\dot{x}, \qquad \ddot{z} = -n^2 z. \quad (1)$$

With state $\mathbf{x} = [x, y, z, \dot{x}, \dot{y}, \dot{z}]^\top$ the solution is $\mathbf{x}(t) = \Phi(t)\,\mathbf{x}(0)$ with the closed-form STM

$$\Phi(t) = \begin{bmatrix} \Phi_{rr} & \Phi_{rv} \\ \Phi_{vr} & \Phi_{vv} \end{bmatrix}, \quad (2)$$

$$\Phi_{rr} = \begin{bmatrix} 4-3c & 0 & 0 \\ 6(s-nt) & 1 & 0 \\ 0 & 0 & c \end{bmatrix}, \qquad \Phi_{vv} = \begin{bmatrix} c & 2s & 0 \\ -2s & 4c-3 & 0 \\ 0 & 0 & c \end{bmatrix},$$

$$\Phi_{rv} = \tfrac{1}{n}\begin{bmatrix} s & 2(1-c) & 0 \\ -2(1-c) & 4s-3nt & 0 \\ 0 & 0 & s \end{bmatrix},$$

$$\Phi_{vr} = n\begin{bmatrix} 3s & 0 & 0 \\ -6(1-c) & 0 & 0 \\ 0 & 0 & -s \end{bmatrix},$$

where $s = \sin nt$, $c = \cos nt$. All propagation in this work—including the optimization dynamics—uses $\Phi$ directly, so there is no numerical integration error. Impulsive burns are modeled as instantaneous velocity increments, propagated piecewise: at a burn time $t_b$ the state is advanced to $t_b$, incremented by $B\Delta\mathbf{v}$ with $B = [\,0_{3\times3};\, I_3\,]$, and the segment epoch reset. A pure along-track offset $[0, y_0, 0, 0, 0, 0]^\top$ is an equilibrium of (1), which is why the unperturbed formation holds its spacing indefinitely. For visualization, the chief is propagated analytically in ECI and relative states are mapped to absolute position via the RIC direction-cosine matrix and the transport theorem.

### 3.2. Two-impulse baseline

The baseline controller returns only the perturbed satellite to its original slot with burns at $t_d$ and $t_f$. Writing the transfer horizon $\tau = t_f - t_d$ and the coasted state at the first burn $\mathbf{x}(t_d) = [\mathbf{r}_0^\top, \mathbf{v}_0^\top]^\top$, the required post-burn velocity and arrival burn are

$$\mathbf{v}_0^+ = \Phi_{rv}^{-1}(\tau)\big[\mathbf{r}_f - \Phi_{rr}(\tau)\,\mathbf{r}_0\big], \quad (3)$$

$$\Delta\mathbf{v}_2 = \mathbf{v}_f - \big[\Phi_{vr}(\tau)\,\mathbf{r}_0 + \Phi_{vv}(\tau)\,\mathbf{v}_0^+\big], \quad (4)$$

with $\Delta\mathbf{v}_1 = \mathbf{v}_0^+ - \mathbf{v}_0$. Note that $\Phi_{rv}$ is singular whenever $n\tau$ is a multiple of $\pi$ (the in-plane block), so the baseline's cost diverges as the horizon approaches half-period multiples—a structural weakness the free-burn-time SCP formulation does not share. The baseline also carries no collision-avoidance logic.

### 3.3. SCP formation recovery

Figure 2 summarizes the planner. We discretize the controlled window $[t_d, t_f]$ into $N$ steps of length $\Delta t$ and allow an impulse $\Delta\mathbf{v}_k^{(i)}$ for every satellite $i$ at every node $k = 0, \ldots, N$ ("burn as many times as necessary"). Burn times on a grid are the convex surrogate for continuous burn-time optimization, which is nonconvex (the burn time enters the STM nonlinearly); as $\Delta t \to 0$ the grid solution approaches the continuous impulsive optimum, and the group-lasso objective below keeps the selected burn set finite and sparse regardless of grid density. Path constraints, by contrast, are imposed continuously in time (sub-sampled between nodes), as described below. For a fixed slot assignment $\sigma$ and recovery time $t_f$, the subproblem is

$$\begin{aligned} \min_{\Delta\mathbf{v},\mathbf{x},s,m} \; & \sum_i \sum_k \big\|\Delta\mathbf{v}_k^{(i)}\big\|_2 + w_{\mathrm{bal}}\, m + w_s \textstyle\sum s \quad (5) \\ \text{s.t. } & \mathbf{x}_{k+1}^{(i)} = \Phi(\Delta t)\big(\mathbf{x}_k^{(i)} + B\,\Delta\mathbf{v}_k^{(i)}\big), \\ & \mathbf{x}_N^{(i)} + B\,\Delta\mathbf{v}_N^{(i)} = \mathbf{x}_{\mathrm{slot},\sigma(i)}, \\ & \big\|\Delta\mathbf{v}_k^{(i)}\big\|_2 \le \Delta v_{\max}, \quad \textstyle\sum_k \big\|\Delta\mathbf{v}_k^{(i)}\big\|_2 \le \min(v_{\mathrm{fuel}},\, m), \\ & \hat{\mathbf{n}}^{ij}(\tau)^\top\big(\mathbf{r}^{(i)}(\tau) - \mathbf{r}^{(j)}(\tau)\big) \ge d_{\min} + \epsilon - s^{ij}(\tau), \\ & \qquad s^{ij}(\tau) \ge 0 \quad \forall\, i < j, \;\; \tau \in \mathcal{T}_c, \end{aligned}$$

where $m$ is an epigraph variable for the maximum per-satellite fuel use. The objective combines three terms. The sum of $\ell_2$ impulse norms is the total constellation $\Delta v$; as a group lasso it drives most candidate impulses to exactly zero, so the solver selects the number of burns. The balancing term $w_{\mathrm{bal}}\, m$ penalizes concentrating fuel use on one vehicle, since mission life ends when the first satellite exhausts its budget. The slack penalty $w_s \sum s$ is an exact penalty that preserves subproblem feasibility during early iterations. Time of

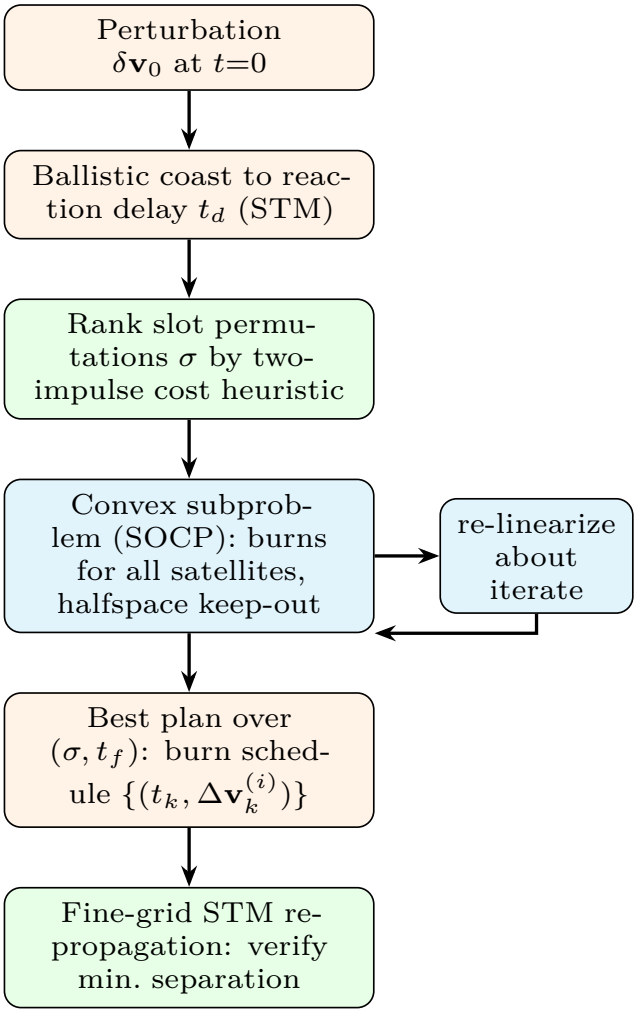



Figure 2: SCP recovery pipeline. The exact CW STM propagates the uncontrolled coast; slot permutations are ranked by a closed-form heuristic; each candidate assignment is solved as a sequence of SOCPs whose only linearized element is the keep-out constraint, imposed at densely sub-sampled times along every coast segment; the winning plan is re-propagated on a fine grid to confirm the realized minimum separation.

flight enters through an outer search: candidate $t_f$ values are each solved and compared by $J = \Delta v_{\text{tot}} + w_{t_f} t_f$, trading fuel against time out of formation.

Continuous-time collision enforcement: The coasting state between burn nodes is $\mathbf{x}(t_k + \tau) = \Phi(\tau)\big(\mathbf{x}_k + B\,\Delta\mathbf{v}_k\big)$, which is affine in the decision variables, so the keep-out constraint can be imposed at arbitrary times without changing the problem class. We impose it on the dense sample set $\mathcal{T}_c$, every $\Delta t/6$ along every coast segment (exact STM interpolation, no added burn nodes), with a margin $\epsilon = 10\,\text{m}$ padding $d_{\min}$ against the residual excursion between adjacent samples. The trajectory therefore cannot pass through a keep-out sphere between burn nodes, and a fine-grid STM re-propagation of the executed plan confirms the realized minimum separation.

Linearization: The CW dynamics in (5) are exactly linear, so the pairwise keep-out constraint is the only non-convexity. At each constraint time $\tau$, the previous iterate supplies the relative (line-of-sight) position $\bar{\mathbf{p}}^{ij}(\tau) = \bar{\mathbf{r}}^{(i)}(\tau) - \bar{\mathbf{r}}^{(j)}(\tau)$, and we constrain the new iterate to the supporting halfspace of the keep-out sphere with normal $\hat{\mathbf{n}}^{ij}(\tau) = \bar{\mathbf{p}}^{ij}(\tau)/\|\bar{\mathbf{p}}^{ij}(\tau)\|_2$—the hyperplane tangent to the sphere where the line of sight exits it, perpendicular to that line of sight. This halfspace is entirely outside the sphere, so the linearization is a conservative inner approximation of the true feasible set: every iterate satisfying it is collision-free at every imposed sample, and no trust region is needed for constraint validity [9, 10]. Iteration re-aims the halfspace normal to recover optimality; we iterate from a ballistic reference until the total $\Delta v$ changes by less than $10^{-3}\,\text{m/s}$.

Slot assignment: Any permutation $\sigma$ of the satellites over the original slots restores the required spacing. We rank all $n!$ permutations by the summed closed-form two-impulse cost (3) of sending each satellite to its candidate slot, then solve (5) exactly for the best few. For $n$=3 this heuristic-plus-exact search adds negligible cost.

### 3.4. Codebase

All components are implemented in our Python package cwsim: the STM and propagation (dynamics), analytic chief orbit and RIC/ECI frames (orbits, frames), formation containers (satellites), the simulation driver (simulation), the SCP planner and baseline (scp), and a dual-frame interactive 3D visualization with a textured Earth (visualization, built on PyVista [14]). Convex subproblems are modeled in CVXPY [12] and solved with Clarabel [13]. The project proposal targeted MATLAB; we moved to Python for the open-source conic-solver and 3D-rendering ecosystem.

## 4. Simulation Setup and Verification

Scenario: Three homogeneous satellites in an equally spaced along-track formation ($\pm 1\,\text{km}$) about a circular 550 km, 51.6° chief orbit (period 95.6 min). At $t$=0 the middle satellite (Sat-2) receives an unexpected $\delta\mathbf{v}_0$; no corrective burns are allowed until the reaction delay $t_d$. Table 1 lists the scenario and solver settings. We assume CW dynamics, perfect noise-free

Table 1: Scenario and Solver Settings.

| Parameter | Setting |
|---|---|
| Chief orbit | circular, $h_0 = 550\,\text{km}$, $i = 51.6^\circ$ |
| Orbit period $T$ | 95.6 min; $n = \sqrt{\mu/a_0^3}$ |
| Formation | 3 satellites, $\pm 1\,\text{km}$ along-track |
| Keep-out radius $d_{\min}$ | 200 m (pairwise) |
| Keep-out sampling $\mathcal{T}_c$ | every $\Delta t/6$ (STM interpolation) |
| Collision margin $\epsilon$ | 10 m pad on $d_{\min}$ |
| Per-burn limit $\Delta v_{\max}$ | 0.5 m/s |
| Fuel budget $v_{\text{fuel}}$ | 5 m/s per satellite |
| Burn grid $\Delta t$ | 150 s (sweep); 90 s (demo) |
| Balance weight $w_{\text{bal}}$ | 0.1 |
| Slack penalty $w_s$ | 1 (m/s)/m (exact penalty) |
| Time weight $w_{t_f}$ | $2 \times 10^{-5}$ (m/s)/s (demo) |
| SCP convergence | $\lvert\Delta(\Delta v_{\text{tot}})\rvert < 10^{-3}$ m/s |
| Assignments solved | best 2–3 of 3! (heuristic-ranked) |
| Solver | CVXPY + Clarabel (SOCP) |
| Recovery time $t_f$ | $2.15\,T$ (sweep); traded (demo) |

Table 2: Numerical verification of the dynamics stack.

| Check | Max error |
|---|---|
| STM vs. RK4 (1.5 orbits), position | $4.1 \times 10^{-12}$ m |
| STM semigroup $\Phi(t_1{+}t_2) = \Phi(t_2)\Phi(t_1)$ | machine precision |
| Two-impulse arrival position error | $7.1 \times 10^{-15}$ m |
| RIC $\leftrightarrow$ ECI round-trip | $< 10^{-9}$ (all states) |
| Along-track equilibrium drift (5000 s) | $< 10^{-9}$ m |

state knowledge, and perfect impulsive execution, per the project problem statement.

Verification: Because every downstream result rests on the STM, we verify the dynamics stack directly (Table 2): the STM matches RK4 integration of (1) at the $10^{-12}$ m level over 1.5 orbits, satisfies the semigroup property $\Phi(t_1{+}t_2) = \Phi(t_2)\Phi(t_1)$ and $\Phi(0) = I$, the two-impulse law (3) achieves its target to machine precision, the RIC$\leftrightarrow$ECI mapping round-trips exactly, and the along-track offset equilibrium is preserved.

## 5. Experiments, Results, and Discussion

Metrics: We report total constellation $\Delta v$ (sum of all impulse magnitudes over all satellites), the chosen slot assignment, the number of burns the group-lasso objective selects, and the fine-grid minimum pairwise separation over the full window $[0, t_f]$, including the uncontrolled delay. Both architectures are evaluated at identical $(t_d, t_f)$ for a fair comparison.

Headline scenario: For $\delta\mathbf{v}_0 = [0.05, 0.10, 0.03]$ m/s on Sat-2 and $t_d = 0.25\,T$, the baseline returns Sat-2 to its slot for 2.27 m/s. The SCP planner instead swaps Sat-1 and Sat-2 (Fig. 1) for a total of 0.30 m/s—an 87% reduction—splitting the effort 0.12/0.18 m/s between the two movers while Sat-3 holds station, with 379 m minimum separation against the 200 m keep-out. With the time weight active, the outer search also selects the shortest candidate recovery time, since the fuel saved by waiting longer was smaller than the time penalty.
Trade study: Tables 3 and 4 and Figures 3–5 sweep the perturbation direction and magnitude and the reaction delay at fixed $t_f = 2.15\,T$ (chosen so that no swept horizon $t_f - t_d$ lands on a half-period multiple, where the baseline diverges; see Sec. 3.2).

### 5.1. Analysis and discussion

We organize the analysis by perturbation direction, since the CW dynamics decouple the three channels into qualitatively different responses, then discuss the delay trade, collision safety, and solver behavior.

Along-track: re-slotting always wins. An along-track $\delta v$ changes the effective semi-major axis and produces secular drift—the expensive, dangerous channel, and the core of the project's hypothesis. In every along-track case in Tables 3 and 4, the SCP planner chooses to swap Sat-2 with the neighbor it is drifting toward rather than reverse the drift, cutting cost by 23–61% (e.g. 0.270 vs. 0.695 m/s at $t_d = 0.5\,T$; the smallest margin, 23%, occurs at the one-orbit delay, where the accumulated drift has made even the swap expensive). The mechanism is visible in Figure 3 (middle row): Sat-2 rides its natural drift loop into the adjacent slot while Sat-1 flies a short cooperative transfer to the vacated center—two cheap maneuvers aligned with the dynamics instead of one expensive maneuver against them. This directly answers the project question: when the perturbation is along-track, leveraging the drift through re-ordering is uniformly cheaper than restoring the original slot order.

Radial: recovery at half the perturbation. A radial $\delta v$ excites the bounded 2×1 relative ellipse with no secular term. The SCP cost is $\approx |\delta v|/2$ at every delay (e.g. 0.050 for a 0.100 m/s kick): with freely chosen burn times the planner cancels the oscillation at a favorable phase, exploiting the 2× along-track-to-radial control authority of the Coriolis coupling. The baseline, whose burn times are pinned to $t_d$ and $t_f$, pays up to 9× more when those times land at unfavorable phases (0.452 at $t_d = 0.1\,T$) and only approaches the SCP cost when the delay happens to align the phase.

Cross-track: cost exactly $|\delta v|$. The out-of-plane channel is a decoupled harmonic oscillator, so the minimum

Table 3: Total constellation $\Delta v$ [m/s] vs. reaction delay ($|\delta \mathbf{v}_0| = 0.1\,\mathrm{m/s}$, $t_f = 2.15\,T$). Parenthesized superscript $^{\mathrm{sw}}$ marks SCP solutions that chose to re-slot (swap Sat-1↔Sat-2). The baseline maneuvers only Sat-2.

| Perturbation | Method | $t_d = 0.1\,T$ | $t_d = 0.25\,T$ | $t_d = 0.5\,T$ | $t_d = 1.0\,T$ |
|---|---|---|---|---|---|
| Radial | Baseline | 0.452 | 0.308 | 0.102 | 0.100 |
| | SCP | 0.050 | 0.050 | 0.050 | 0.056 |
| Along-track | Baseline | 0.379 | 0.464 | 0.695 | 0.387 |
| | SCP | $0.167^{\mathrm{sw}}$ | $0.205^{\mathrm{sw}}$ | $0.270^{\mathrm{sw}}$ | $0.299^{\mathrm{sw}}$ |
| Cross-track | Baseline | 0.452 | 0.308 | 0.100 | 0.100 |
| | SCP | 0.100 | 0.100 | 0.100 | 0.100 |

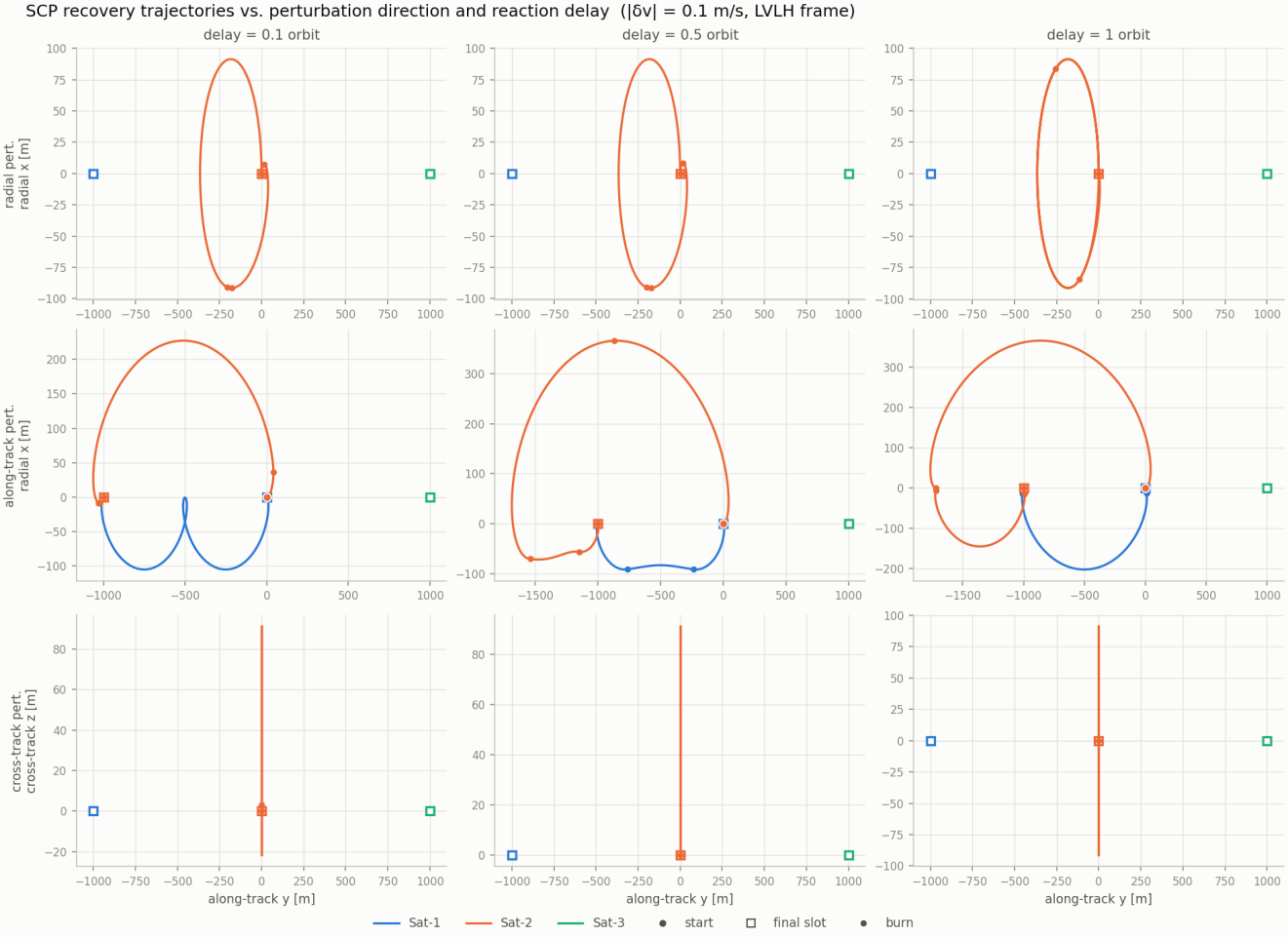



Figure 3: SCP recovery trajectories in the RIC frame as the perturbation direction (rows) and reaction delay (columns) vary, at $|\delta \mathbf{v}_0| = 0.1\,\mathrm{m/s}$. Radial perturbations (top) produce a bounded 2×1 in-plane ellipse that the planner closes for $\approx |\delta v|/2$. Along-track perturbations (middle) accumulate secular drift that grows with delay; in every case the planner swaps Sat-2 (orange) with Sat-1 (blue) rather than reversing the drift, and at longer delays Sat-1 flies a cooperative transfer to the vacated center slot. Cross-track perturbations (bottom) excite a decoupled out-of-plane oscillation cancelled for exactly $|\delta v|$.

recovery cost is exactly the perturbation magnitude—cancel the oscillation at a node crossing—and the SCP solution achieves 0.100 m/s at every delay, usually with a single burn. The phase-locked baseline again pays up

Table 4: Total constellation $\Delta v$ [m/s] vs. perturbation magnitude ($t_d = 0.25\,T$, $t_f = 2.15\,T$).

| Perturbation | Method | 0.05 m/s | 0.1 m/s | 0.2 m/s |
|---|---|---|---|---|
| Radial | Baseline | 0.154 | 0.308 | 0.616 |
| | SCP | 0.025 | 0.050 | 0.100 |
| Along-track | Baseline | 0.232 | 0.464 | 0.928 |
| | SCP | $0.162^{\text{sw}}$ | $0.205^{\text{sw}}$ | $0.409^{\text{sw}}$ |
| Cross-track | Baseline | 0.154 | 0.308 | 0.616 |
| | SCP | 0.050 | 0.100 | 0.200 |

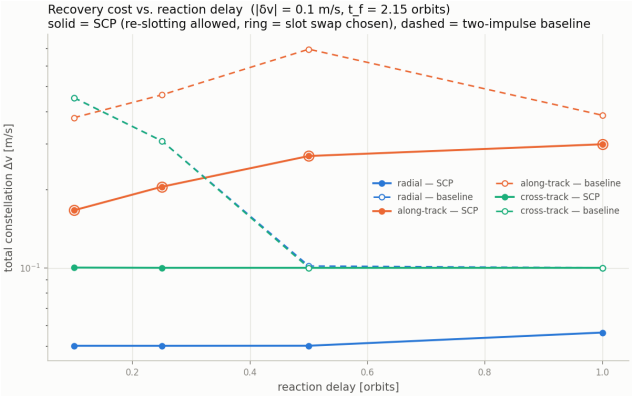



Figure 4: Total constellation $\Delta v$ vs. reaction delay ($|\delta\mathbf{v}_0| = 0.1$ m/s, log scale). Solid: SCP (ring = slot swap chosen); dashed: two-impulse baseline. SCP cost is flat in delay for the bounded radial and cross-track responses, and grows slowly for along-track, while the phase-locked baseline varies strongly with burn timing.

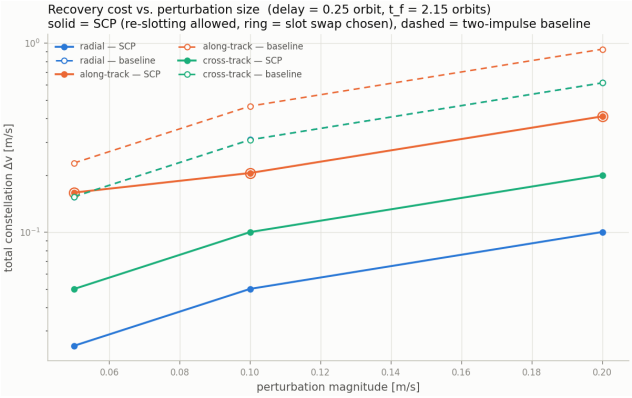



Figure 5: Total constellation $\Delta v$ vs. perturbation magnitude ($t_d = 0.25\,T$, log scale). Costs scale essentially linearly with $|\delta\mathbf{v}_0|$ for both methods; SCP maintains a 1.5–6× advantage across the range.

to 4.5× more at short delays. Neither radial nor cross-track cases ever trigger re-slotting: the responses are bounded, so there is no drift to exploit.

Reaction delay. Delay matters only where drift accumulates. Along-track SCP cost grows from 0.167 to 0.299 m/s as the delay grows from $0.1\,T$ to $1.0\,T$—waiting is strictly costly when the satellite is drifting away—while radial and cross-track costs are essentially flat because the uncontrolled motion is bounded. The baseline's non-monotone delay dependence is dominated by burn phasing rather than by the physics of the perturbation, a further argument for burn-time-free planning.

Collision safety. The keep-out constraint is active, not decorative: the along-track drift carries Sat-2 directly through Sat-1's position. The baseline, with no collision logic, dips to 192.6 m minimum separation in one swept case—inside the 200 m keep-out. With the keep-out imposed at every $\Delta t/6$ along each coast segment plus the 10 m margin, every SCP plan's fine-grid minimum separation stays strictly outside the sphere ($\geq 210.6$ m across the sweep); the one case where the constraint binds pays only 0.003 m/s more than node-only enforcement did.

Solver behavior. Across the sweep, subproblems solve in seconds and the SCP loop converges in 2 iterations when the ballistic reference is already collision-feasible, and in 6–8 iterations when the initial halfspace linearization forces a large detour that subsequent relinearizations relax. The exact-penalty slack is nonzero only on the first iteration of hard cases and returns to zero thereafter, consistent with the conservative inner approximation guaranteeing feasible iterates. The group-lasso objective selects 2–9 burns per plan without any integer variables.

## 5.2. Limitations

Our conclusions inherit the CW assumptions: linearized relative dynamics about a circular chief, no $J_2$, drag, or navigation and execution errors, and impulsive burns whose times live on a dense grid (continuous burn-time optimization is non-convex; the grid solution approaches it as $\Delta t \to 0$). The keep-out constraint is sampled densely in time with a margin rather than imposed exactly over continuous intervals, and is non-convex, so SCP returns locally optimal plans with no global certificate—the assignment search mitigates, but does not eliminate, this. The slot-assignment heuristic could in principle rank the true optimum below the permutations solved exactly, and the study covers a three-satellite formation; the formulation scales to $n$ satellites, but the $n!$ assignment enumeration would

need replacement (e.g. by a linear assignment relaxation) for large $n$.

## 6. Conclusion and Future Work

We presented cwsim, a modular CW formation-flying simulator built on the exact closed-form 3-DOF STM, and used it to compare two recovery architectures for a perturbed three-satellite along-track formation: a closed-form two-impulse return-to-slot baseline and an SCP planner that coordinates burns across all satellites with re-slotting, keep-out spheres, thrust limits, fuel budgets, and fuel balancing. Across perturbation directions, magnitudes, and reaction delays, SCP reduces total constellation $\Delta v$ by up to 89% (median $\approx$56%); for along-track perturbations the optimizer always re-slots, confirming the project hypothesis that the natural drift should be leveraged rather than reversed. The study also surfaced two practical warnings for operators: the two-impulse law diverges near half-period transfer horizons, and without explicit collision constraints it can violate keep-out margins that the convex planner honors by construction.

Future work includes validating the plans against nonlinear two-body and $J_2$-perturbed dynamics, formal continuous-interval collision certificates (bounding the excursion between constraint samples), Monte Carlo robustness under navigation and execution errors, larger constellations with relaxation-based assignment, and a receding-horizon variant that replans as tracking errors accumulate.

## Contributions and Acknowledgements

Rahul Ayanampudi: CW dynamics and STM implementation, two-impulse baseline, SCP formulation and implementation, trade-study design, simulator and visualization, report writing.

Zeb Barry:

Julian Brown:

Generative AI usage. We used AI tools to supplement portions of this work. AI assisted with generating the initial codebase scaffolding for the simulator, visualization, and solver modules, and with LaTeX table and figure formatting. We maintained oversight of the formulation and verified the dynamics stack against independent numerical integration (Table 2); all problem definitions, parameter choices, reported metrics, and conclusions were reviewed and verified by us.

## References


[1] W. H. Clohessy and R. S. Wiltshire. Terminal guidance system for satellite rendezvous. Journal of the Aerospace Sciences, 27(9):653–658, 1960.

[2] G. W. Hill. Researches in the lunar theory. American Journal of Mathematics, 1(1):5–26, 1878.

[3] D. A. Vallado. Fundamentals of Astrodynamics and Applications. Microcosm Press, 4th edition, 2013.

[4] H. Schaub and J. L. Junkins. Analytical Mechanics of Space Systems. AIAA Education Series, 4th edition, 2018.

[5] K. T. Alfriend, S. R. Vadali, P. Gurfil, J. P. How, and L. Breger. Spacecraft Formation Flying: Dynamics, Control and Navigation. Elsevier, 2010.

[6] NASA Ames Research Center. Starling: Swarm satellite technology demonstration. https://www.nasa.gov/mission/starling/, 2023.

[7] B. Açıkmeşe and S. R. Ploen. Convex programming approach to powered descent guidance for Mars landing. Journal of Guidance, Control, and Dynamics, 30(5):1353–1366, 2007.

[8] P. Lu and X. Liu. Autonomous trajectory planning for rendezvous and proximity operations by conic optimization. Journal of Guidance, Control, and Dynamics, 36(2):375–389, 2013.

[9] D. Morgan, S.-J. Chung, and F. Y. Hadaegh. Model predictive control of swarms of spacecraft using sequential convex programming. Journal of Guidance, Control, and Dynamics, 37(6):1725–1740, 2014.

[10] Y. Mao, M. Szmuk, and B. Açıkmeşe. Successive convexification of non-convex optimal control problems and its convergence properties. In IEEE Conference on Decision and Control, pages 3636–3641, 2016.

[11] S. Boyd and L. Vandenberghe. Convex Optimization. Cambridge University Press, 2004.

[12] S. Diamond and S. Boyd. CVXPY: A Python-embedded modeling language for convex optimization. Journal of Machine Learning Research, 17(83):1–5, 2016.

[13] P. J. Goulart and Y. Chen. Clarabel: An interior-point solver for conic programs with quadratic objectives. arXiv preprint arXiv:2405.12762, 2024.

[14] C. B. Sullivan and A. Kaszynski. PyVista: 3D plotting and mesh analysis through a streamlined interface for the Visualization Toolkit (VTK). Journal of Open Source Software, 4(37):1450, 2019.